

TECHNICAL DRAWING SET

Interconnection

Wiring of the mini-PC, camera, audio, relay board, garage door, indicators and door sensor.

Reference installation: Lenovo ThinkCentre M73 Tiny · Ubuntu Server 26.04 LTS · ONVIF PTZ camera · Novoferm Novomatic 200 garage door operator. 100 % on-premises processing.

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Date of issue	2026-09-30
Author	Thierry Gayet <thierry.gayet@labworks.fr>
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RECORD OF REVISIONS

Rev	Date	Description
A	2026-09-29	Initial release (French edition).
B	2026-09-30	US English edition. Content aligned with the current repository (web front end with Anubis, MQTT, remote syslog, monitoring, NTP, TLS certificate lifecycle, camera power and end-of-life data). Contents with page numbers, glossary, bibliography and index added.

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1 SCOPE AND SAFETY SUMMARY

This document gives the wiring of every element around the Lenovo ThinkCentre M73 Tiny mini-PC: networks, PoE camera, audio chain, USB relay board, garage door operator, LED indicators and door sensor, with the allocation of the ports of the Tiny. Everything connected to the door, the indicators and the sensor is **extra-low voltage**; Jarvis never switches 230 V.

DANGER

Electric shock hazard. Disconnect the Novomatic 200 from the mains before opening it or wiring terminal F. Never connect 230 V to the relay board, to the FT232RL or to the LED circuit.

The anti-crushing safety remains the job of the door operator itself (photocells and force detection): never bypass them.

CAUTION

The voltage present on terminal F (external pulse input) is not documented by Novoferm. Measure it with a multimeter before wiring and confirm that it is a low-voltage, potential-free input.

CAUTION

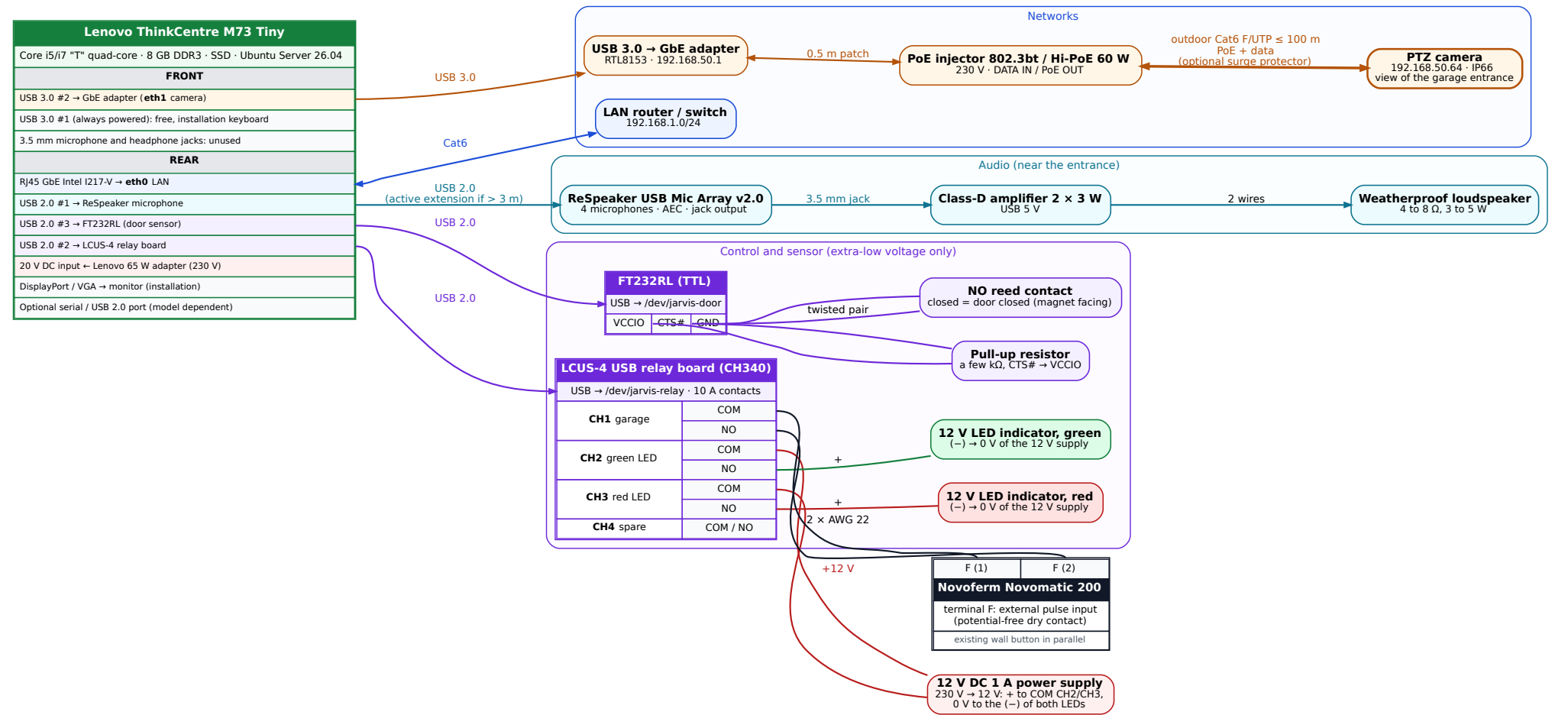
A single pulse cycles through open / stop / close. Without the door sensor, “open” sent to an open door closes it: fit the reed contact.

NOTE

The PoE injector and the 12 V supply are the only mains-powered items of the installation besides the mini-PC adapter. Use enclosed, CE-certified units.

2 WIRING DIAGRAM

power LAN camera / PoE audio USB I/O, relay, sensor door command



3 PORT ALLOCATION OF THE TINY

Port	Connected to	Linux name	Configuration / remarks
Front USB 3.0 #1 (always powered)	free	—	Installation keyboard / USB stick.
Front USB 3.0 #2	USB → GbE adapter	eth1 (e.g. enx...)	Camera network 192.168.50.0/24. USB 3.0 is required for 1 Gbit/s.
Rear RJ45	LAN router / switch	eth0 (enp0s25...)	Web UI, SSH, integrations.
Rear USB 2.0 #1	ReSpeaker USB Mic Array v2.0	ALSA "ReSpeaker 4 Mic Array"	audio.input_device and audio.output_device = this device.
Rear USB 2.0 #2	LCUS-4 relay board (CH340 1a86:7523)	/dev/jarvis-relay	udev rule deploy/udev/99-jarvis.rules.
Rear USB 2.0 #3	FT232RL (0403:6001)	/dev/jarvis-door	door_sensor: {backend: serial_cts, device: /dev/jarvis-door}
Rear DisplayPort / VGA	Monitor	—	Installation only.
Front 3.5 mm jacks	unused	—	An analog microphone picks up poorly at a distance; the ReSpeaker is preferred.
Optional serial port (model dependent)	alternative to the FT232RL	/dev/ttyS0	RS-232 levels: wire the reed between DTR (pin 4) and CTS (pin 8), not to GND.

4 WIRING DETAILS

4.1 Novomatic 200 garage door (CH1)

Relay CH1 COM ————
 Relay CH1 NO ————
 terminal F "external pulse input"
 of the Novomatic 200 (Novoferm manual, fig. 13)
 (in parallel with the existing wall button)

- Dry contact: the relay closes the contact for **500 ms** (`hardware.pulse_ms`), like a button press, with a 5 s cooldown between two pulses.
- Cable: 2 × AWG 22 per the Novoferm manual (fig. 13a); Wago 221 lever terminals accept 0.14 to 4 mm².
- One pulse = open / stop / close, in that order: hence the door sensor.

4.2 LED indicators (CH2, CH3)

+12 V → COM CH2 — NO CH2 → (+) green LED (–)
 +12 V → COM CH3 — NO CH3 → (+) red LED (–) → 0 V of the 12 V supply

4.3 Door sensor (reed contact)

FT232RL VCCIO —[pull-up, a few kΩ]—
 FT232RL CTS# ———— NO reed contact
 FT232RL GND ———— (closed = magnet facing)

- Door closed → reed closed → CTS# pulled to ground → `cts = True` → "closed" (`active_means_closed: true`, the default).
- The FT232R internal pull-up is only **about 200 kΩ** (DS_FT232R v2.16): with a long cable, add an **external pull-up of a few kΩ to VCCIO** and filtering (IO-9).
- Input threshold 1.0 to 1.5 V; never apply a voltage outside –0.5 V to VCC + 0.5 V.
- **With an RS-232 (DB9) cable**, CTS tied to GND does not work (0 V is not a valid level): wire the reed between **DTR** (pin 4, asserted by pyserial when the port is opened) and **CTS** (pin 8).
- Twisted pair, away from 230 V cables; 20 to 30 m are no problem for this slow signal.

4.4 Lengths and limits

- Ethernet / PoE: **100 m max.** between the injector and the camera, UV-resistant outdoor cable, drip loop.
- USB 2.0 passive: **5 m max.**; beyond, an active extension or a USB-over-RJ45 extender.
- PoE power: 802.3bt / Hi-PoE 60 W (up to 51 W at the camera); PoE+ 802.3at (30 W) is not enough.

APPENDIX A GLOSSARY

Technical terms and abbreviations used in this document.

802.3af / at / bt

IEEE Power over Ethernet standards: 15.4 W (af, "PoE"), 30 W (at, "PoE+"), 60 to 90 W at the source (bt, "PoE++", classes 5 to 8). The PTZ cameras of the project draw up to 51 W and need 802.3bt or the vendor's Hi-PoE injector.

AEC

Acoustic Echo Cancellation: removal of the loudspeaker signal from the microphone signal. Performed here by the XMOS XVF-3000 chip of the ReSpeaker.

CH340

WCH USB-to-serial converter chip fitted to the LCUS relay boards.

Cooldown

Minimum delay enforced between two pulses of the garage relay (5 s by default).

CTS / DTR

Control lines of a serial link: CTS (Clear To Send) is an input, DTR (Data Terminal Ready) an output. CTS is used as the logic input of the door sensor.

Dry contact (potential-free)

Electrical contact that supplies no voltage: it only closes or opens a circuit powered by the connected equipment. The only type allowed on terminal F of the Novomatic 200.

F/UTP

Twisted-pair cable with an overall foil screen (Foiled / Unshielded Twisted Pairs).

FT232R / FT232RL

FTDI USB-to-serial converter chip with TTL levels; its CTS# pin reads the door sensor. Its internal pull-up (about 200 kΩ) is weak.

Hi-PoE / High PoE

Proprietary high-power PoE (Hikvision, Dahua; "High PoE" at Axis), close to 802.3bt.

Injector (PoE midspan)

Unit that adds power to an Ethernet cable between a network device and a powered device.

IP55 / IP65 / IP66 / IP67

Ingress protection ratings (IEC 60529): 6 = dust-tight; 5 = water jets, 6 = powerful jets, 7 = temporary immersion.

LCUS-4

4-relay board driven over USB through a CH340, serial frame A0 channel state checksum.

Lever terminal block

Screwless connector with a lever (Wago 221 type) for 0.14 to 4 mm² conductors.

NO / NC / COM

Relay terminals: COM common, NO normally open (closed when the relay is energized), NC normally closed.

PoE

Power over Ethernet: electrical power carried by the network cable.

PTZ

Pan-Tilt-Zoom: camera motorized in pan, tilt and zoom.

Pull-up

Resistor that pulls a logic input to the high state when nothing drives it low.

Pulse

Short closure (500 ms) of the relay contact, equivalent to pressing the push button.

Reed switch

Switch with flexible blades operated by a magnet; used as a door opening sensor.

Similarity threshold

Minimum cosine between two embeddings to conclude that they belong to the same person (0.45 for faces).

udev

Linux device manager; its rules give stable names (/dev/jarvis-relay).

Unknown cluster

Automatic grouping of the visits of the same unknown visitor.

APPENDIX B BIBLIOGRAPHY

Further reading. Titles are given as published; ISBNs are those of the edition cited and are omitted when they could not be verified.

B.1 Books in English

1. Åström, Karl Johan; Murray, Richard M. *Feedback Systems: An Introduction for Scientists and Engineers*. Princeton University Press, 2008. ISBN 978-0-691-13576-2.
2. Horowitz, Paul; Hill, Winfield. *The Art of Electronics*. 3rd ed. Cambridge University Press, 2015. ISBN 978-0-521-80926-9.
3. Kerrisk, Michael. *The Linux Programming Interface: A Linux and UNIX System Programming Handbook*. No Starch Press, 2010. ISBN 978-1-59327-220-3.
4. Ward, Brian. *How Linux Works: What Every Superuser Should Know*. 3rd ed. No Starch Press, 2021. ISBN 978-1-7185-0040-2.

B.2 Books in French

1. Blaess, Christophe. *Développement système sous Linux : ordonnancement multitâche, gestion mémoire, communications, programmation réseau*. 5th ed. Eyrolles, 2019. ISBN 978-2-212-67760-7.
2. Rohaut, Sébastien; Banquet, Philippe. *Linux : maîtrisez l'administration du système*. 7th ed. Éditions ENI, 2024. ISBN 978-2-409-04366-6.

B.3 Standards, specifications and papers

1. IEEE Std 802.3bt-2018. *Physical Layer and Management Parameters for Power over Ethernet over 4 pairs*.
2. IEC 60529. *Degrees of protection provided by enclosures (IP Code)*.

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